

NARENDRA KUMAR POLUKA
Senior Data Scientist | AI/ML Engineer
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PROFESSIONAL SUMMARY

- Accomplished Senior Data Scientist with 10+ years of experience delivering enterprise-scale Artificial Intelligence, Machine Learning, Predictive Analytics, Deep Learning, NLP, and Generative AI solutions across Financial Services, Retail Pharmacy, Manufacturing, and Technology domains within the United States and India.
- Extensive hands-on experience designing and deploying AI-powered applications using Python 3.x, Scikit-Learn 1.x, TensorFlow 2.x, PyTorch 2.x, XGBoost 2.x, LightGBM 4.x, and Spark MLlib, supporting business-critical decision-making and operational automation initiatives.
- Proven expertise developing production-grade Generative AI and LLM solutions leveraging OpenAI GPT-4o, Azure OpenAI, Claude 3.x, Gemini 1.5, LangChain, LangGraph, LlamaIndex, RAG architectures, Semantic Search, Vector Databases, and AI Agents.
- Strong experience architecting cloud-native AI platforms on AWS, Azure, and GCP utilizing SageMaker, Bedrock, Azure Machine Learning, Azure OpenAI Service, Vertex AI, Databricks Runtime, and Snowflake for scalable machine learning operations.
- Hands-on expertise building end-to-end MLOps frameworks using MLflow 2.x, Kubeflow, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, and CI/CD automation pipelines to streamline model deployment, governance, and monitoring.
- Extensive background in predictive modeling, fraud detection, customer analytics, recommendation engines, risk assessment, forecasting, anomaly detection, customer segmentation, and real-time decision intelligence using structured and unstructured enterprise data.
- Experienced in designing large-scale data engineering solutions utilizing Apache Spark 3.x, PySpark, Kafka, Airflow, Delta Lake, Databricks, Snowflake, and distributed processing frameworks to support high-volume analytical workloads.
- Specialized in Natural Language Processing (NLP) using Hugging Face Transformers, BERT, RoBERTa, DistilBERT, SpaCy, NLTK, OCR, and document intelligence solutions for automated content extraction, classification, and business process optimization.
- Demonstrated success implementing AI-driven fraud analytics, payment intelligence, customer behavior modeling, demand forecasting, and operational optimization solutions that improved business efficiency, customer experience, and revenue growth.
- Strong experience collaborating with Product Owners, Business Stakeholders, Enterprise Architects, Data Engineers, and Executive Leadership teams to translate complex business challenges into scalable AI and advanced analytics solutions.
- Adept at leading cross-functional teams, mentoring Data Scientists and Machine Learning Engineers, conducting model reviews, establishing AI governance standards, and driving enterprise-wide adoption of modern AI technologies.
- Skilled in Agile/Scrum delivery methodologies, responsible AI practices, model explainability, risk governance, and regulatory compliance while consistently delivering high-impact AI and analytics programs across large enterprise environments.

TECHNICAL SKILLS

Programming Languages	Python 3.x, SQL, Java 17, Scala 2.13, R, Shell Scripting
Generative AI & LLMs	GPT-4o, GPT-4, Claude 3.x, Gemini 1.5, Llama 3, Mistral, LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen

AI/ML Frameworks	Scikit-Learn 1.x, TensorFlow 2.x, PyTorch 2.x, XGBoost 2.x, LightGBM 4.x, CatBoost, Spark MLlib
NLP & Document AI	BERT, RoBERTa, DistilBERT, Hugging Face, SpaCy, NLTK, OCR, Document Intelligence
RAG & AI Search	Retrieval-Augmented Generation (RAG), Semantic Search, Embeddings, Prompt Engineering, AI Agents
Vector Databases	Pinecone, Weaviate, ChromaDB, FAISS, Milvus
Big Data Technologies	Apache Spark 3.x, PySpark, Kafka, Hadoop, Hive, Databricks Runtime, Delta Lake
Cloud Platforms	AWS SageMaker, Bedrock, S3, Lambda, EMR, Azure OpenAI, Azure ML, Azure Databricks, GCP Vertex AI
Data Engineering	Airflow, Snowflake, Databricks, ETL/ELT Pipelines, Data Modeling, Data Warehousing
MLOps & DevOps	MLflow, Kubeflow, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform
Databases	PostgreSQL, SQL Server, Oracle, MySQL, MongoDB, Cassandra, Redis
BI & Visualization	Power BI, Tableau, Looker, Plotly, Matplotlib, Seaborn
Methodologies	Agile, Scrum, SDLC, CI/CD, Responsible AI, Model Governance, Data Governance

PROFESSIONAL EXPERIENCE

Client: Fiserv, Milwaukee, WI

Role: Senior Data Scientist – AI/ML

Duration: Oct 2023 – Present

- Spearheaded the design and deployment of enterprise-scale AI/ML solutions supporting digital payments, fraud prevention, merchant analytics, customer intelligence, and risk management initiatives across Fiserv's financial services ecosystem, processing billions of payment transactions annually.
- Architected advanced machine learning platforms utilizing Python 3.x, Scikit-Learn 1.x, XGBoost 2.x, LightGBM 4.x, TensorFlow 2.x, and PyTorch 2.x to develop predictive models that enhanced fraud detection accuracy and reduced financial risk exposure.
- Designed and implemented real-time fraud detection frameworks leveraging Apache Spark 3.x, Kafka, Databricks Runtime, and feature engineering pipelines capable of analyzing high-volume payment transactions with low-latency scoring capabilities.
- Developed AI-powered risk assessment and transaction monitoring solutions using supervised and unsupervised machine learning algorithms, anomaly detection models, graph analytics, and behavioral profiling techniques that improved fraud investigation efficiency by more than 35%.
- Built customer segmentation, merchant intelligence, and customer lifetime value models utilizing advanced statistical techniques, clustering algorithms, and predictive analytics frameworks to drive targeted marketing strategies and improve customer retention outcomes.
- Led the development of Generative AI-powered financial knowledge assistants utilizing Azure OpenAI, GPT-4o, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), and vector databases to improve internal knowledge management and operational productivity.
- Designed scalable data science pipelines using Databricks, Delta Lake, Apache Airflow, Snowflake, and Azure Data Factory, enabling automated ingestion, transformation, and processing of structured and semi-structured financial datasets.
- Implemented end-to-end MLOps frameworks leveraging MLflow 2.x, Docker, Kubernetes, Azure Machine Learning, GitHub Actions, and CI/CD automation pipelines, significantly reducing model deployment cycles and improving production reliability.

- Developed advanced payment forecasting, transaction volume prediction, and merchant performance analytics models utilizing time-series forecasting, machine learning, and statistical modeling techniques to support strategic business planning initiatives.
- Built NLP-driven document intelligence solutions using Hugging Face Transformers, BERT, SpaCy, OCR technologies, and text analytics frameworks to automate financial document classification, compliance reviews, and operational reporting workflows.
- Collaborated extensively with Product Owners, Fraud Operations teams, Compliance stakeholders, Enterprise Architects, and Executive Leadership to identify AI opportunities and translate complex financial business requirements into scalable analytical solutions.
- Established model governance and Responsible AI frameworks incorporating model explainability, bias detection, drift monitoring, performance tracking, auditability controls, and regulatory compliance requirements supporting enterprise AI adoption.
- Engineered real-time customer analytics solutions utilizing Spark Structured Streaming, Kafka, Azure Event Hubs, and distributed computing technologies to provide actionable insights into transaction behavior, customer engagement, and operational performance metrics.
- Mentored Data Scientists, Machine Learning Engineers, and Analytics professionals by conducting architecture reviews, technical design sessions, model validation activities, and best-practice workshops focused on scalable AI and ML solution development.
- Delivered multiple high-impact AI, Machine Learning, and Generative AI initiatives that improved fraud prevention effectiveness, accelerated operational decision-making, enhanced merchant experiences, optimized payment processing workflows, and generated measurable business value across Fiserv's financial technology platform.

Client: CVS, Irving, TX

Role: Senior Data Scientist

Duration: Aug 2020 – Sep 2023

- Led enterprise-wide Artificial Intelligence, Machine Learning, and Advanced Analytics initiatives supporting healthcare operations, pharmacy services, claims processing, member engagement, population health, and enterprise digital transformation programs.
- Architected and deployed scalable machine learning solutions utilizing Python 3.x, Scikit-Learn 1.x, TensorFlow 2.x, PyTorch 2.x, XGBoost 2.x, and Spark MLlib to improve clinical outcomes, operational efficiency, and business decision-making across healthcare domains.
- Developed predictive models for patient risk stratification, hospital readmission prediction, medication adherence, and chronic disease management using large-scale healthcare claims, pharmacy, and member datasets.
- Designed real-time analytics pipelines using Apache Kafka, Spark Structured Streaming, Databricks Runtime, Delta Lake, and Azure Event Hubs to process millions of healthcare transactions, pharmacy events, and member interactions for near real-time insights.
- Built fraud, waste, and abuse (FWA) detection models leveraging anomaly detection, supervised learning, graph analytics, and behavioral modeling to identify suspicious claims and reduce financial losses.
- Developed demand forecasting and capacity planning models for pharmacy inventory optimization, prescription volume forecasting, workforce planning, and healthcare resource allocation using statistical forecasting and machine learning techniques.
- Collaborated with clinicians, pharmacy operations teams, care management, business stakeholders, product managers, and executive leadership to identify high-impact AI opportunities and deliver data-driven healthcare solutions.
- Built healthcare analytics platforms integrating claims data, electronic health records (EHR), pharmacy data, provider information, and member engagement data to enable predictive insights and operational reporting.

- Designed member segmentation, customer analytics, recommendation systems, and personalized outreach models to improve member engagement, preventive care participation, and healthcare experience.
- Established enterprise MLOps frameworks using MLflow 2.x, Kubeflow, Docker, Kubernetes, Jenkins, GitHub Actions, and Azure DevOps to automate model development, deployment, monitoring, governance, and CI/CD pipelines.
- Developed NLP-based document intelligence solutions utilizing BERT, RoBERTa, SpaCy, Hugging Face Transformers, and OCR technologies to automate extraction of information from clinical notes, provider documents, prior authorization requests, and medical records.
- Built cloud-native AI platforms on Azure and AWS utilizing Azure Machine Learning, Databricks, AWS SageMaker, Snowflake, and scalable data lake architectures to support enterprise healthcare analytics and machine learning workloads.
- Implemented model governance, explainability, and monitoring frameworks incorporating SHAP, model drift detection, bias evaluation, performance monitoring, and regulatory compliance practices aligned with HIPAA and enterprise AI governance standards.
- Mentored cross-functional teams of Data Scientists, Data Engineers, Analytics Consultants, and Machine Learning Engineers by establishing technical standards, conducting architecture reviews, and promoting best practices for enterprise AI development.
- Successfully delivered high-impact AI, predictive analytics, healthcare intelligence, fraud detection, and machine learning initiatives that improved patient outcomes, optimized pharmacy operations, enhanced claims processing efficiency, reduced operational costs, and accelerated CVS Health's digital transformation initiatives.

Client: Fifth Third Bank, Evansville, IN

Role: Senior Data Scientist

Duration: Jan 2019 – Jul 2020

- Led the design and delivery of Machine Learning and Predictive Analytics solutions supporting retail banking, consumer lending, fraud detection, credit risk, customer analytics, and digital banking initiatives.
- Developed scalable predictive models using Python, Scikit-Learn, XGBoost, TensorFlow, Spark MLlib, and statistical techniques to improve credit decisioning, customer engagement, and operational efficiency.
- Built credit risk, loan default prediction, and customer propensity models using behavioral, transactional, and financial data to optimize lending strategies and portfolio performance.
- Designed customer segmentation, churn prediction, recommendation, and cross-sell models to enhance personalized banking experiences and increase product adoption.
- Engineered real-time analytics pipelines using Apache Spark, Kafka, Hadoop, Hive, and AWS to process large-scale transaction and customer interaction data.
- Implemented fraud detection and Anti-Money Laundering (AML) models leveraging anomaly detection, supervised learning, and risk scoring techniques to strengthen fraud prevention and regulatory compliance.
- Collaborated with risk management, lending, compliance, marketing, and business stakeholders to translate analytical insights into enterprise-scale business solutions.
- Developed NLP solutions using SpaCy, NLTK, and Word2Vec to analyze customer feedback, service interactions, and financial documents for operational and customer experience improvements.
- Built executive dashboards and self-service reporting solutions using Tableau, Power BI, SQL, and AWS Redshift to monitor portfolio performance, fraud trends, customer metrics, and operational KPIs.
- Established automated MLOps pipelines using MLflow, Docker, Jenkins, Git, and cloud-native technologies to streamline model deployment, monitoring, and lifecycle management.
- Partnered with Data Engineering teams to improve data quality, optimize ETL pipelines, and develop trusted analytical datasets supporting enterprise reporting and AI initiatives.

- Performed feature engineering, statistical analysis, model validation, and explainability using SHAP to improve predictive performance and ensure transparent, production-ready models.
- Delivered high-impact machine learning solutions that reduced fraud risk, improved lending decisions, enhanced customer retention, optimized operational performance, and supported enterprise-wide data-driven decision-making.

Client: Samsung SDS Infotech Pvt Ltd, Gurgaon, India

Role: Data Scientist

Duration: Sep 2016 – May 2018

- Developed enterprise-scale Machine Learning and Advanced Analytics solutions supporting supply chain management, logistics optimization, customer intelligence, operational analytics, and digital transformation initiatives for global business operations and technology clients.
- Designed and implemented predictive modeling frameworks using Python, R, SQL, Scikit-Learn, XGBoost, and statistical analysis techniques to improve forecasting accuracy, operational visibility, and data-driven decision-making across multiple business domains.
- Built large-scale data processing and analytics pipelines leveraging Apache Spark, Hadoop, Hive, Sqoop, and distributed computing frameworks to process and analyze high-volume structured and unstructured enterprise datasets.
- Developed demand forecasting, inventory planning, and logistics optimization models utilizing machine learning algorithms, time-series forecasting techniques, and supply chain analytics methodologies to improve operational efficiency and resource utilization.
- Collaborated closely with business analysts, product teams, solution architects, and client stakeholders to understand business requirements and deliver scalable analytics solutions aligned with strategic organizational goals.
- Engineered customer segmentation and behavioral analytics solutions using clustering techniques, classification models, and predictive intelligence frameworks to support personalized engagement strategies and customer experience improvements.
- Built real-time and batch data integration pipelines using Spark SQL, Kafka, ETL frameworks, and enterprise data warehouses to support operational reporting, analytics, and business intelligence initiatives.
- Performed extensive exploratory data analysis, feature engineering, statistical modeling, and hypothesis testing to identify business trends, operational inefficiencies, and growth opportunities across multiple enterprise functions.
- Developed Natural Language Processing (NLP) applications utilizing NLTK, SpaCy, text mining, sentiment analysis, and document classification techniques to analyze customer communications, service records, and operational documentation.
- Designed anomaly detection and operational monitoring solutions leveraging machine learning algorithms and statistical models to proactively identify performance deviations, process bottlenecks, and operational risks.
- Created executive dashboards and business intelligence solutions using Tableau, Power BI, SQL Server Reporting Services (SSRS), and Excel, enabling stakeholders to monitor KPIs, operational performance, and strategic business metrics.
- Worked extensively with relational and non-relational databases including Oracle, SQL Server, MySQL, MongoDB, and Hadoop ecosystems to support enterprise analytics, reporting, and data management requirements.
- Supported cloud adoption and big data modernization initiatives by integrating analytical workloads with scalable infrastructure platforms, improving data accessibility, processing performance, and analytical capabilities.
- Participated in Agile software development processes including sprint planning, backlog grooming, user story refinement, solution design discussions, and stakeholder reviews to ensure successful project execution and delivery.

- Delivered multiple high-impact analytics and machine learning solutions that improved forecasting accuracy, enhanced supply chain visibility, optimized business operations, reduced manual processes, and generated measurable value for enterprise clients and business stakeholders.

Client: Blue Star Infotech Ltd, Mumbai, India

Role: Associate Data Analyst

Duration: Jul 2014 – Aug 2016

- Analyzed large-scale business, operational, customer, and transactional datasets using SQL, Python, Excel, and statistical analysis techniques to support enterprise reporting, business intelligence initiatives, and data-driven decision-making processes across multiple departments.
- Developed automated reporting solutions and KPI dashboards utilizing SQL Server, Tableau, Power BI, Excel VBA, and advanced reporting frameworks, significantly reducing manual reporting efforts while improving data visibility and stakeholder access to critical business metrics.
- Collaborated with business users, project managers, technical teams, and functional stakeholders to gather analytical requirements, define reporting objectives, and deliver actionable insights supporting operational and strategic business initiatives.
- Performed extensive data profiling, cleansing, transformation, validation, and reconciliation activities across multiple enterprise systems to improve data quality, ensure reporting accuracy, and establish trusted datasets for analytics and business intelligence solutions.
- Assisted in developing predictive analytics and forecasting models using Python, R, regression analysis, statistical modeling, and machine learning fundamentals to support trend forecasting, business planning, and operational performance improvement initiatives.
- Designed and maintained ETL workflows utilizing SQL, Informatica PowerCenter, SSIS, and data integration tools to extract, transform, and load enterprise data from multiple heterogeneous source systems into centralized reporting repositories.
- Conducted exploratory data analysis, root cause investigations, variance analysis, and trend identification exercises to uncover operational bottlenecks, performance gaps, and process improvement opportunities across various business functions.
- Created detailed executive reports, analytical scorecards, operational dashboards, and management reporting packages that enabled leadership teams to monitor organizational performance, evaluate business outcomes, and make informed decisions.
- Supported enterprise data warehouse initiatives by assisting with dimensional modeling, schema design, metadata documentation, data mapping, and governance activities to improve reporting consistency and analytical scalability.
- Applied statistical techniques including hypothesis testing, correlation analysis, variance analysis, forecasting, and descriptive statistics to generate meaningful insights and support evidence-based business decision-making processes.
- Worked closely with software development teams, database administrators, QA teams, and business stakeholders to troubleshoot data issues, optimize SQL queries, enhance reporting performance, and ensure successful delivery of analytical solutions.
- Participated in data migration, system modernization, and application enhancement projects involving data extraction, mapping, transformation, reconciliation, testing, and validation activities across enterprise platforms.
- Assisted in implementing self-service reporting and business intelligence solutions that improved user adoption, reduced reporting turnaround times, and enabled business users to access critical information independently.
- Contributed to Agile project delivery by supporting requirement gathering, documentation preparation, user acceptance testing, sprint activities, stakeholder communications, and production support efforts throughout project lifecycles.